

From interval palettes to dense nonsumsets in prime fields

Project proof artifact for AIM Problem 2.7

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Abstract

For a prime p , let $M(p)$ be the greatest cardinality of a subset of $\mathbb{F}_p$ which is not $B + B$ for any $B \subseteq \mathbb{F}_p$, with equal summands allowed. We prove

$$M(p) = p - p^{1/2+o(1)}.$$

The proof combines Noga Alon’s universal large-set theorem with the interval-palette theorem in Shouqiao Wang’s pinned proposed solution of Erdős Problem 788. We transfer two interval palettes to consecutive charts of $\mathbb{Z}_p$, bound every possible self-sumset root by an independence number, and use a strict finite count to choose an unrealized output. All imported statements and new reductions are identified explicitly. Wang’s theorem is machine-checked but recent and not peer-reviewed, so the result is presented as a candidate proof for expert review rather than as a community-accepted resolution.

1 Introduction

The problem, recorded as AIM Problem 2.7 by B. Green [1], asks for the largest size of a subset of a prime field which is not an unrestricted self-sumset. Theorem 1 determines the exponent of the missing part: the extremal nonsumsets have $p^{1/2+o(1)}$ missing elements. The argument is short once one has a suitable low-score graph on the additive group, but the passage from the available interval graph to the cyclic group and the requirement of an *exact* sumset output both need justification.

Contribution and dependency boundary. There are two imported results. Alon’s published theorem gives the lower bound on the number of missing elements. Wang’s interval-palette theorem gives graphs whose palette size plus independence number is $N^{1/2+o(1)}$. The new part of this paper consists of the normalization of Wang’s theorem, the two-chart cyclic transfer, and the exact-output deletion argument. The mathematical deduction is complete relative to the two stated inputs. Appendix A records separately why the recent Wang input is treated as machine-checked evidence rather than as an already accepted literature result.

Proof roadmap. The proof has five steps.

1. Alon’s theorem shows that deleting at most $\frac{1}{4000} \sqrt{p/\log p}$ elements always leaves a self-sumset.
2. Wang’s theorem supplies an interval graph whose palette size e and independence number r have sum $p^{1/2+o(1)}$ after rescaling.

3. Two consecutive interval charts cover $\mathbb{Z}_p$ and transfer these palettes to a cyclic graph Γ_E without increasing the score $H = e + r$ beyond $p^{1/2+o(1)}$.
4. If $(B + B) \cap E = \emptyset$, then the distinct elements of B form an independent set in Γ_E , so every possible root has $|B| \leq r$.
5. There are more $4H$ -element deletion sets outside E than there are roots of size at most r . One deletion therefore produces an exact output which is not $B + B$ for any B .

The lower and upper estimates for the number of missing elements then have the same exponent.

2 Statement and notation

For $B \subseteq \mathbb{F}_p$, define the unrestricted self-sumset

$$B + B = \{b_1 + b_2 : b_1, b_2 \in B\}.$$

The empty root is allowed, and $b_1 = b_2$ is allowed. Define

$$M(p) = \max\{|A| : A \subseteq \mathbb{F}_p \text{ and } A \neq B + B \text{ for every } B \subseteq \mathbb{F}_p\}$$

and put $d(p) = p - M(p)$. Thus $d(p)$ is the strict complement threshold, not the inclusive threshold used in some earlier papers.

Theorem 1 (AIM Problem 2.7). *As p tends to infinity through the primes,*

$$d(p) = p^{1/2+o(1)} \quad \text{and} \quad M(p) = p - p^{1/2+o(1)}.$$

All logarithms are natural. We identify $\mathbb{F}_p$ with the additive group $\mathbb{Z}_p$ and with its standard representatives $\{0, \dots, p-1\}$.

3 Two imported results

The lower estimate follows from the following exact inclusive statement.

Theorem 2 (Alon, Theorem 2.1 [2]). *For every sufficiently large prime p , if $F \subseteq \mathbb{Z}_p$ and*

$$|F| \leq \frac{1}{4000} \sqrt{\frac{p}{\log p}},$$

then $\mathbb{Z}_p \setminus F = A + A$ for some $A \subseteq \mathbb{Z}_p$.

We next state exactly the consequence of Wang's interval theorem which is used here. For $N \geq 2$ and $Q \subseteq \{1, \dots, 2N-3\}$, let H_Q be the simple graph on

$$[N]_0 = \{0, 1, \dots, N-1\}$$

in which distinct x, y are adjacent precisely when $x + y \in Q$.

Theorem 3 (Interval-palette theorem [3]). *There is an absolute constant $C > 0$ such that, for every sufficiently large integer N , there is $Q_N \subseteq \{1, \dots, 2N-3\}$ satisfying*

$$|Q_N| + \alpha(H_{Q_N}) \leq (N+1)^{1/2+\Delta_N}, \quad \Delta_N = C \left(\frac{\log \log(N+1)}{\log(N+1)} \right)^{1/3}.$$

For completeness, this is the exact normalization of the cited result. Wang writes

$$I_n = (n, 2n) \cap \mathbb{N} = \{n + 1, \dots, 2n - 1\}$$

and proves both the min-max identity

$$f(n) = \min_{D \subseteq J_n} (|D| + \alpha(G_D)),$$

with the minimum unchanged when D is restricted to attainable sums of two distinct elements of I_n , and the upper bound

$$f(n) \leq n^{1/2 + C(\log \log n / \log n)^{1/3}}.$$

Set $n = N + 1$, translate I_n by $-(n + 1)$, and translate every palette sum by $-(2n + 2)$. The vertex set becomes $[N]_0$, the attainable distinct sums become exactly $\{1, \dots, 2N - 3\}$, and the graph and its independence number are preserved. This gives Theorem 3 without an asymptotic change of parameters.

4 Transfer to a cyclic palette

For $E \subseteq \mathbb{Z}_p$, let Γ_E be the simple graph on $\mathbb{Z}_p$ in which distinct x, y are adjacent exactly when $x + y \in E$.

Lemma 4 (Root-size reduction). *For any $E, B \subseteq \mathbb{Z}_p$, if $(B + B) \cap E = \emptyset$, then B is independent in Γ_E . In particular,*

$$|B| \leq \alpha(\Gamma_E).$$

Proof. If distinct $x, y \in B$ were adjacent, then $x + y \in (B + B) \cap E$, contrary to the hypothesis. Thus no two distinct vertices of B are adjacent. Notice that equal summands remain in $B + B$; the conclusion only needs the distinct-pair edges of the simple graph. $\square$

Lemma 5 (Two-chart transfer). *For every sufficiently large odd prime p , there is $E \subseteq \mathbb{Z}_p$ such that, with*

$$e = |E|, \quad r = \alpha(\Gamma_E), \quad H = e + r,$$

one has

$$2\sqrt{p} - 1 \leq H \leq p^{1/2 + \eta_p}$$

for a sequence $\eta_p \rightarrow 0$.

Proof. Put

$$N_0 = \frac{p+1}{2}, \quad N_1 = \frac{p-1}{2}, \quad a_0 = 0, \quad a_1 = N_0,$$

and partition $\mathbb{Z}_p$ into the two standard intervals

$$V_i = a_i + \{0, \dots, N_i - 1\} \quad (i = 0, 1).$$

Choose Q_{N_i} from Theorem 3 and define

$$E_i = \{2a_i + q \pmod{p} : q \in Q_{N_i}\}, \quad E = E_0 \cup E_1.$$

Because $Q_{N_0} \subseteq \{1, \dots, p-2\}$ and $Q_{N_1} \subseteq \{1, \dots, p-4\}$, reduction modulo p is injective on each palette. Cross-palette collisions are allowed and only decrease $|E|$. Thus

$$e \leq |Q_{N_0}| + |Q_{N_1}|.$$

Let R be independent in Γ_E . For fixed i , define

$$R_i = \{u \in [N_i]_0 : a_i + u \in R\}.$$

If distinct $u, v \in R_i$ satisfied $u + v \in Q_{N_i}$, then

$$(a_i + u) + (a_i + v) \equiv 2a_i + u + v \pmod{p} \in E_i \subseteq E,$$

contradicting independence. Hence R_i is independent in $H_{Q_{N_i}}$. The two charts partition $\mathbb{Z}_p$, so

$$r \leq \alpha(H_{Q_{N_0}}) + \alpha(H_{Q_{N_1}}).$$

Let $\varepsilon_p = \max_i \Delta_{N_i}$ and $\eta_p = \varepsilon_p + \log 2 / \log p$. Equations (4) and (4), together with Theorem 3, give

$$H \leq \sum_{i=0}^1 (N_i + 1)^{1/2 + \Delta_{N_i}} \leq 2p^{1/2 + \varepsilon_p} = p^{1/2 + \eta_p}.$$

Since $N_i \rightarrow \infty$, we have $\eta_p \rightarrow 0$.

For the lower estimate, each vertex of Γ_E has degree at most e : for fixed x and fixed $s \in E$, at most one neighbor can satisfy $x + y = s$. A graph of maximum degree e has an independent set of size at least $p/(e + 1)$ by greedy coloring. Therefore

$$H = e + r \geq e + \frac{p}{e + 1} = (e + 1) + \frac{p}{e + 1} - 1 \geq 2\sqrt{p} - 1.$$

This argument uses only distinct-pair edges, so no diagonal convention has been changed. $\square$

5 The exact-output deletion

Lemma 6 (Strict finite counting gap). *With e, r, H, η_p as in Lemma 5, for every sufficiently large p ,*

$$3 \leq H, \quad 4H \leq p - e,$$

and

$$\binom{p - e}{4H} > \sum_{j=0}^r \binom{p}{j}.$$

Proof. Take p large enough that

$$\eta_p \leq \frac{1}{16}, \quad \frac{\log 8}{\log p} \leq \frac{1}{16}, \quad 5p^{9/16} \leq p.$$

Lemma 5 gives

$$2\sqrt{p} - 1 \leq H \leq p^{1/2 + \eta_p} \leq p^{9/16}.$$

Hence $H \geq 3$ eventually and, because $e \leq H$, $e + 4H \leq 5p^{9/16} \leq p$.

For integers $0 < k \leq n$,

$$\binom{n}{k} = \prod_{i=0}^{k-1} \frac{n - i}{k - i} \geq \left(\frac{n}{k}\right)^k,$$

because $(n - i)/(k - i) \geq n/k$ is equivalent to $i(n - k) \geq 0$. Apply this with $n = p - e$ and $k = 4H$. Since $e \leq H \leq p^{9/16} \leq p/2$ eventually,

$$\log \binom{p - e}{4H} \geq 4H \log \frac{p - e}{4H} \geq 4H \log \frac{p}{8H}.$$

Using (5) and $H \leq p^{9/16}$, the last logarithm is at least $\frac{3}{8} \log p$, and therefore

$$\binom{p-e}{4H} \geq p^{3H/2}.$$

On the other hand,

$$\sum_{j=0}^r \binom{p}{j} \leq \sum_{j=0}^r p^j \leq p^{r+1} \leq p^{H+1} \leq p^{4H/3} < p^{3H/2},$$

where $H+1 \leq 4H/3$ follows from $H \geq 3$. Combining this strict inequality with (5) proves (6). $\square$

Lemma 7 (Unrealized output). *For every sufficiently large prime p , there is $T \subseteq \mathbb{Z}_p \setminus E$ with $|T| = 4H$ such that*

$$A_p = \mathbb{Z}_p \setminus (E \cup T)$$

is not $B + B$ for any $B \subseteq \mathbb{Z}_p$. Moreover,

$$|\mathbb{Z}_p \setminus A_p| \leq 5p^{1/2+\eta_p}.$$

Proof. Let $\mathcal{T}$ be the family of all $4H$ -element subsets of $\mathbb{Z}_p \setminus E$. By Lemma 6,

$$|\mathcal{T}| = \binom{p-e}{4H} > \sum_{j=0}^r \binom{p}{j}.$$

Call $T \in \mathcal{T}$ realized if

$$B + B = \mathbb{Z}_p \setminus (E \cup T).$$

for some $B \subseteq \mathbb{Z}_p$. Equation (5) implies $(B + B) \cap E = \emptyset$. Lemma 4 therefore gives $|B| \leq r$. There are at most the right-hand side of (5) such roots. In addition, a fixed root determines at most one deletion set, because (5) forces

$$T = (\mathbb{Z}_p \setminus (B + B)) \setminus E.$$

Consequently (5) leaves some T unrealized. The corresponding A_p is a nonsumset. Finally, $E \cap T = \emptyset$ and $e \leq H$, so

$$|\mathbb{Z}_p \setminus A_p| = e + 4H \leq 5H \leq 5p^{1/2+\eta_p}.$$

Equal summands cause no gap: they remain part of (5), while distinct pairs provide the necessary independence condition and (5) uses the full sumset. $\square$

6 Proof of the main theorem

Proof of Theorem 1. Set

$$q_p = \left\lfloor \frac{1}{4000} \sqrt{\frac{p}{\log p}} \right\rfloor.$$

If $S \subseteq \mathbb{Z}_p$ has $|S| \geq p - q_p$, then its complement satisfies the inclusive hypothesis of Theorem 2; hence S is a self-sumset. No nonsumset has size at least $p - q_p$, and integrality gives

$$M(p) \leq p - q_p - 1, \quad d(p) \geq q_p + 1 > \frac{1}{4000} \sqrt{\frac{p}{\log p}}.$$

Lemma 7 supplies a nonsumset A_p with complement at most $5p^{1/2+\eta_p}$. Hence

$$d(p) = p - M(p) \leq p - |A_p| \leq 5p^{1/2+\eta_p}.$$

Taking logarithms in (6) and (6) yields

$$\frac{1}{2} - \frac{\log \log p}{2 \log p} - \frac{\log 4000}{\log p} < \frac{\log d(p)}{\log p} \leq \frac{1}{2} + \eta_p + \frac{\log 5}{\log p}.$$

Both outer expressions tend to $1/2$. Therefore $d(p) = p^{1/2+o(1)}$, and the defining identity $M(p) = p - d(p)$ gives $M(p) = p - p^{1/2+o(1)}$. $\square$

A Verification and evidence boundary

The imported Wang repository was pinned at commit

d28713ac8245ca86a686b8c67370a8d19d81b242.

Its Lean 4.27 development used Mathlib commit

a3a10db0e9d66acbebf76c5e6a135066525ac900

and completed all 7,936 local build jobs. The final theorem depends only on `propext`, `Classical.choice`, and `Quot.sound`; the audited source contains no `sorry`, `axiom`, or `admit`. The interval normalization above was checked against both the paper and the formal definitions.

The new cyclic transfer and deletion proof passed a separate QED xhigh structural audit, a detailed step-by-step audit, deterministic finite falsification checks, and final verdict `DONE`. These checks establish the repository’s proof claim relative to the pinned imported theorem. They do not establish novelty, peer review, or community acceptance. In particular, the source itself is titled a proposed solution and the public Erdős problem page remained marked open at the audit date.

References

- [1] E. Croot and V. F. Lev (collectors), *Problems Presented at the Workshop on Recent Trends in Additive Combinatorics*, AIM workshop list, Problem 2.7 (B. Green), 2004. <https://aimath.org/WWN/additivecomb/additivecomb.pdf>.
- [2] N. Alon, *Large sets in finite fields are sumsets*, J. Number Theory 126 (2007), 110–118. <https://web.math.princeton.edu/~nalon/PDFS/sumset.pdf>.
- [3] S. Wang, *A Proposed Solution to Erdős Problem 788*, pinned repository artifact, commit d28713ac8245ca86a686b8c67370a8d19d81b242, 2026. <https://github.com/ShouqiaoW/erdos/tree/d28713ac8245ca86a686b8c67370a8d19d81b242/788>.